

Project: Number Guessing Game

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Table Of Contents

Title:	3
Description:	3
Features:	3
Code:	3
Conclusion:	5

Title:

Number Guessing Game 🎯

Description:

The Number Guessing Game is a Python-based interactive program where the computer randomly selects a secret number between 1 and 10. The player has three attempts to guess the correct number, receiving feedback after each guess. If the guess is wrong, the game provides helpful hints (such as whether the number is even/odd or greater/less than 5) to guide the player.

Features:

- Random secret number between 1–10.
- Hints after wrong guesses (even/odd, greater/less than 5).
- Replay option to play multiple rounds.
- Input validation (prevents crashes on invalid input).
- User-friendly messages with clear feedback.

Code:

Python:

```
import random
def number_guessing_game():
    secret_number = random.randint(1, 10) # Random number between 1 and 10
    attempts_left = 3

    print("\n Welcome to the Number Guessing Game!")
    print("I am thinking of a number between 1 and 10.")
    print("Can you guess it? You have 3 attempts.\n")
```

```

while attempts_left > 0:
    print("Attempts left:", attempts_left)

    # Input validation
    try:
        guess = int(input("Enter your guess: "))
    except ValueError:
        print(" Invalid input! Please enter a number.")
        continue

    if guess == secret_number:
        print(" Congratulations! You guessed the correct number!")
        break
    elif guess < secret_number:
        print("Too low! Try a higher number.")
        if secret_number % 2 == 0:
            print(" Hint: The secret number is even.")
        else:
            print(" Hint: The secret number is odd.")
    else:
        print("Too high! Try a lower number.")
        if secret_number > 5:
            print(" Hint: The secret number is greater than 5.")
        else:
            print(" Hint: The secret number is less than or equal to 5.")

    attempts_left -= 1

else:
    print("Game Over! The secret number was:", secret_number)
def main():
    while True:
        number_guessing_game()
        replay = input("\n Do you want to play again? (yes/no): ").strip().lower()
        if replay != "yes":
            print("Thanks for playing! Goodbye!")
            break
main()

```

Conclusion:

This project demonstrates the use of loops, conditionals, input validation, and modular programming in Python. It is simple yet effective for learning fundamental programming concepts and is enhanced with hints and replay functionality to improve user experience.